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Optimal sizing and techno-economic analysis of energy- and cost-efficient standalone multi-carrier microgrid

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ABSTRACT

The objective of this study is optimal sizing and techno-economic analysis of an energy- and cost-efficient standalone multi-carrier microgrid (SMCMG). This SMCMG is with the integration of photovoltaic-thermal (PVT), wind turbine (WT), micro-turbine (MT), electrical energy storage (EES), thermal energy storage (TES), and backup natural gas boiler (GB) units. For this aim, a long-term planning approach is put forward, taking into account battery degradation and reliability considerations. Total annual cost (TAC) is served as the objective function of the optimization problem, which covers investment, replacement, fuel, and operation and maintenance costs. A rule-based energy management system (EMS), which prioritizes the application of renewable energy (RE) resources, is introduced to administrate the SMCMG operation and maximize RE utilization. To solve the optimization problem, a robust metaheuristic optimization algorithm, called evolutionary particle swarm optimization (E-PSO), is employed, and the results are validated. The simulations are conducted based on real data from Rafsanjan, Iran.

The superiority of the E-PSO algorithm is demonstrated. Various techno-economic analyses are conducted, and different configurations for SMCMG are studied. The results signify the competence of the proposed configuration for SMCMG from economic, energy efficiency, reliability, and environmental points of view.

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1. Introduction

Integration of distributed generation (DG) micro-sources and energy storage systems can construct a small-scale power grid so-called microgrid (MG) to meet the local loads, which can be operated in both grid-connected and standalone modes. MGs offer numerous benefits such as higher resiliency, reliability, energy efficiency, power quality, to name just a few. They are also envisioned as a new opportunity for the utilization of renewable energy (RE) resources. Among all RE resources, the combination of solar and wind energy has drawn more attention due to their complementary characteristics; however, their intermittent nature prompts for utilization of energy storage systems to meet the time-varying local loads [1].

In this regard, the selection of the best energy mix for the MG and optimal sizing of each component is indispensable to simultaneously ensure the reliable power supply and economic viability of the system. Optimal sizing of a grid-connected MG is less complicated, since the grid plays the role of the backup unit and guarantees the reliability of the system. But in a standalone MG, the reliability of the system must be realized within its boundaries, and this worsens the difficulty and complexity of the problem [2].

In this regard, numerous researches have addressed the optimal sizing of standalone MGs relying on RE resources and storage mechanisms to serve local electrical loads, and different optimization techniques have been employed [3]. Although these systems are environmentally beneficial, reliability and economic dilemmas exist in their utilization.

Recently, the integration of multiple energy carriers, for instance, electricity, hydrogen, natural gas (NG), heating, etc. has been proposed as a solution to levelize demand. It has been accepted that integration of various energy carriers into an optimized framework can open up new opportunities to progress toward a cost-effective, energy-efficient, reliable, and eco-friendly

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energy system [4]. The optimal design and operation of such energy systems with different resources and equipment is a complicated challenge. Hence, they provide an attractive avenue for research and development [5].

An MG with multiple energy carriers is known as a multi-carrier microgrid (MCMG). In this area, numerous studies have underlined the integration of RE resources and combustion-based technologies in standalone MCMG (SMCMG), to address the MCMG with electricity and NG energy carriers [6–9], and for solving the optimization problem, different methods such as particle swarm optimization (PSO) [6], genetic algorithm (GA) [7], harmony search algorithm (HSA) [10], etc. have been investigated. It can be realized that integration of RE and combustion-based technologies results in a lower cost of generation [10], higher system reliability [7], and higher resiliency [11]. Regarding MCMGs based on electricity and hydrogen energy carriers [12], has investigated the optimal sizing of an SMCMG including PV panel, battery, electrolyzer, hydrogen tank (HT), and fuel cell (FC) to fulfill the local electrical loads, and the PSO algorithm has been employed to solve the optimization problem [12]. Moreover [13], has compared the performance of different artificial intelligence techniques for optimal sizing of an SMCMG including PV, wind turbine (WT), and FC. Despite the benefits of the noted MCMGs, meeting the thermal demands needed separate infrastructure, leading to lower overall energy efficiency, utilization, and cost-effectiveness. In other words, the heat produced by the fuel-based micro-sources should be recovered and utilized.

For MCMGs with electrical and thermal loads, combined heat and power (CHP) system is an appropriate solution, which can significantly improve operational energy efficiency. On this subject, some studies have investigated the combination of RE and non-RE resources to meet electrical and thermal loads simultaneously. For instance Ref. [14], has assessed the performance of several grid-connected MCMGs composed of PV panels, solar thermal (ST), and internal combustion engines (ICEs). Further, the optimal sizing of an MCMG including CHP micro-source, gas boiler (GB), and thermal energy storage (TES), connected to the upstream networks of electricity and NG has been investigated in Ref. [15]. It can be inferred from these works that the optimal sizing and configuration selection greatly affects the economic viability and service reliability of the MCMG. A bi-level optimization approach has been presented in Ref. [16] for optimal sizing and operation of a grid-connected MCMG, including CHP unit, PV panel, and GB unit beside the electrical energy storage (EES) and TES with predetermined capacities considering costs of investment, operation, and maintenance. The GA algorithm has been used to find the optimal size of the CHP, PV and GB units, and the mixed-integer nonlinear programming technique has been applied to guarantee the optimum operation of the system. The author of [17] has presented a methodology for optimal design of a grid-connected MCMG including PV panel, ST system, micro-turbine (MT), battery and TES. To solve the optimization problem, the PSO algorithm and evolutionary algorithm have been used and compared. A mixed integer framework for optimal sizing and operation of three different multi-energy systems including PV, WT, gas turbine, EES, and TES have been studied and compared in Ref. [18]. The optimal sizing of discussed MCMGs with thermal and electrical loads have been based on the grid-connected operation.

The existing literature includes a few studies that consider optimal sizing and techno-economic analysis of SMCMG including CHP application. The performance of a PV, FC, electrolyzer, HT energy system has been studied in Ref. [19] for a small remote load. In that study, the HT has been utilized as the single energy storage system; this may limit the flexibility and functionality of the system and increase dump energy. Moreover, the hydrogen storage subsystem (electrolyzer, HT and fuel cell) is a complicated technology with lower efficiencies in comparison to TES. Further, the reliability of such standalone system is not favorable, since backup units for thermal and electrical loads have not been accounted for. In Ref. [20], the optimal sizing of an SMCMG including FC, PV, ST system, battery, TES, and GB, to serve the local thermal and electrical loads, has been studied. In that work, FC has been considered as the main CHP micro-source for the SMCMG and GA has been used for optimization but without any validation of results. Also [21], has come up with the idea of consideration of security constraint in optimal sizing of SMCMG with electrical and thermal loads, including PV panel, FC, MT, ICE, ST system, and GB. The results show that although utilization of RE resources may seem to be sensible, in a minimum cost design with security constraints, utilization of PV panel among several dispatchable CHP micro-sources may be unaffordable in an SMCMG with both electrical and thermal loads. Therefore, it can be concluded that CHP based RE resources are the micro-sources that must be utilized in SMCMG with electrical and thermal loads.

A photovoltaic-thermal (PVT) produces electrical and thermal energy simultaneously. Further, the total efficiency (electrical efficiency + thermal efficiency) of the PVT is higher than the sum of the electrical efficiency of the PV panel and thermal efficiency of the ST system. Hence, it can be utilized as an RE based-CHP unit to serve local electrical and thermal loads [22]. For a standalone application, a study on optimal sizing of the PVT panel along with TES and EES has been carried out for meeting residential electrical and thermal loads [23]. In that study, it has been demonstrated, to ensure the fulfillment of the loads, the connection to the upstream electricity and NG networks is indispensable for that system. For optimization, GA has been used without the validation of the results with the results of other methods.

Considering the published literature, it can be said that the determination of the best energy mix for SMCMGs with electrical and thermal loads requires further research with the focus on optimal sizing and techno-economic analysis. In recent works of authors, greenhouse gas (GHG) emission-free standalone combined cooling, heating, and power (CCHP) [24] and CHP [25] systems including PVT, WT along with storage mechanisms have been studied. But, those systems suffered from significant low energy efficiency, because a huge amount of produced electrical and thermal energy, namely more than 30%, was dumped. Moreover, for a higher level of reliability, utilization of backup units for thermal and electrical loads seems to be vital in those standalone systems. The noted issues along with lack of proper flexibility and functionality can lead to restriction of utilization of such systems for SMCMGs with heavy loads. It is expected that incorporation of a dispatchable CHP micro-source can serve our purposes in the SMCMG in terms of more flexibility and reliability along with lower dump energy and thus higher energy efficiency. In this regard, it is expected that the combination of two different CHP technologies, namely PVT panel (RE-based CHP micro-source) and fossil fuel-based CHP micro-source is the solution. Meanwhile, among dispatchable CHP micro-sources, MT can use different fuels, emits less pollution, requires less maintenance, and is more reliable. Thus, it is the best alternative for standalone applications, although it is not GHG emission-free. Furthermore, utilization of a backup GB unit for thermal energy seems effective for more flexible operation as well as lower investment cost. Also, it is anticipated that the benefits in energy efficiency, cost, and reliability criteria far outweigh the GHG emission of the SMCMG. Further, the GHG emission of the SMCMG is alleviated by the energy management system (EMS) of the SMCMG by giving priority to RE resources. On the other hand, over the years, the performance of the battery degrades and its capacity drops. It is expected that this degradation can affect the

performance of the SMCMG and therefore must be considered in the sizing procedure.

The objective of this study is optimal sizing and techno-economic analysis of an energy- and cost-efficient SMCMG including PVT panel, WT, MT, EES, TES, and backup GB unit. For this aim, a long-term planning approach is put forward, taking into account investment, replacement, fuel, and operation and maintenance costs along with reliability considerations. Further, the influence of battery degradation is also integrated into the optimization procedure. For energy management in hourly time intervals, a rule-based EMS, which considers giving priority to RE resources is introduced to administrate the SMCMG operation and maximize RE utilization.

To justify the proposed configuration, the impacts of each component of the SMCMG on total annual cost (TAC), service reliability, and total produced dump energy of the SMCMG is investigated. A cost versus reliability analysis is also conducted, and different configuration alternatives are compared in this regard. Several sensitivity analyses on important parameters, such as fuel price, interest rate, and optimal capacities of EES and TES are also conducted to determine the impacts of their variations on the techno-economic aspects of the system. Furthermore, environmental benefits resulting from the operation of the proposed SMCMG is addressed.

On the other hand, all the previous works emphasize the importance of the employed optimization technique in the economic efficiency of the system and the validity of results. In this area, metaheuristic optimization algorithms are commonly used because of their good performance, and also their function is independent of the objective function of the problem. However, the downside of these optimization methods is the possibility of getting trapped in local optimal points. To overcome this issue, a robust and newly developed algorithm referred to as evolutionary particle swarm optimization (E-PSO) is used and evaluated through sensitivity and comparative analyses.

The main contributions of this study are summarized as follows:

(i) Optimal sizing of a new configuration of SMCMG with the integration of dispatchable and non-dispatchable CHP technologies, WTs, and hybrid storage systems.
(ii) Considering the influence of battery degradation on the results of the SMCMG sizing.
(iii) Comprehensive techno-economic analysis of different configuration alternatives for SMCMG.
(iv) Employment of a robust metaheuristic optimization technique for optimization of SMCMGs.

The remainder of this study is structured as follows. Next, the SMCMG architecture and modeling is presented. Problem formulation is outlined in section 3. The optimization method is presented in section 4. Simulation results and techno-economic analyses are investigated in section 5, and the concluding remarks are addressed in section 6. A schematic diagram of the simulation and the connection between different sections of the manuscript is drawn in Fig. 1.

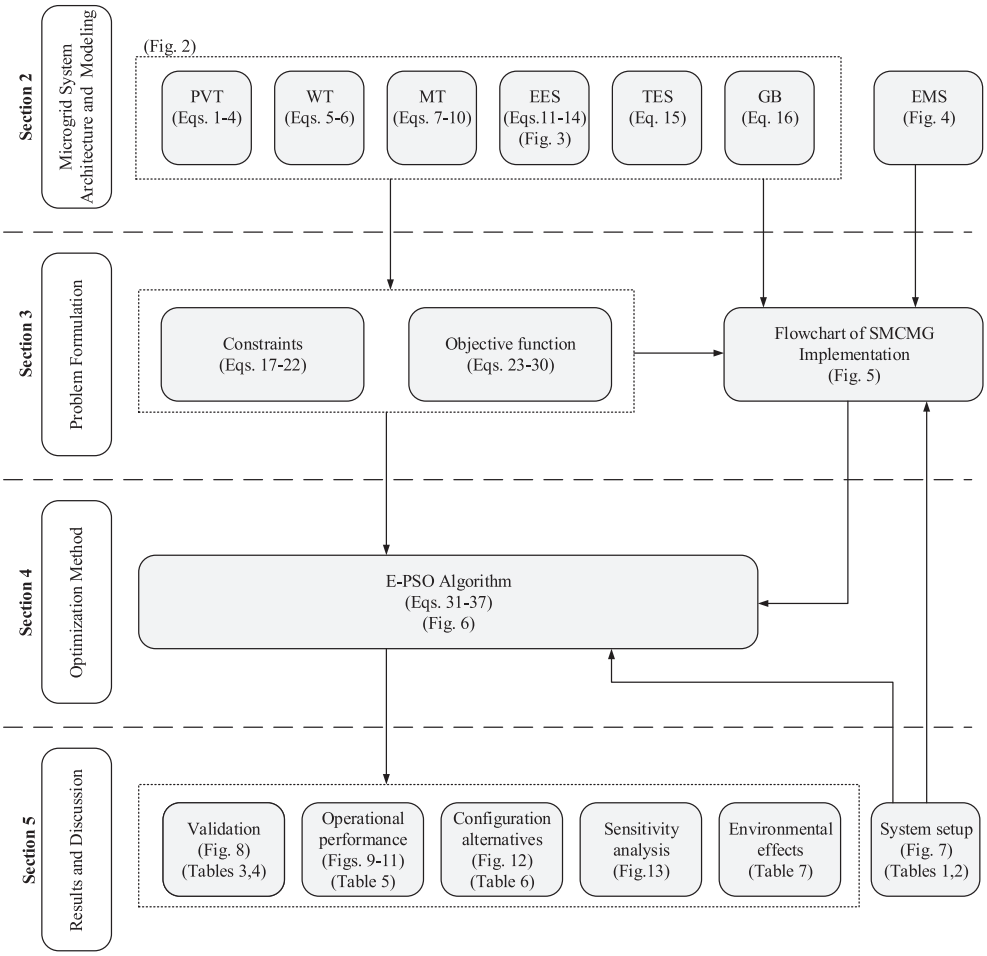


Fig. 1. Schematic diagram of the simulation and connection between different sections.

2. Microgrid system architecture and modeling

Fig. 2 illustrates the studied SMCMG which provides electrical and thermal energy to the loads through electrical and thermal hubs. The PVT panel as the RE based-CHP unit is used to simultaneously generate electrical and thermal energy, WTs are incorporated as the complement of electricity generation of the PVT panel, and MT as the dispatchable CHP micro-source is employed to improve the flexibility and coordination of the SMCMG by means of fulfilling electrical load in case of battery depletion while its high-temperature exhaust gas is used to accommodate the thermal load. The storage systems including a heat storage tank and a battery bank as the TES and EES, respectively, are utilized to coordinate the imbalance between supply and load, and a supplementary GB is used as a backup source of thermal load. Further, utilization of a dump load is indispensable to prevent the battery from overcharging.

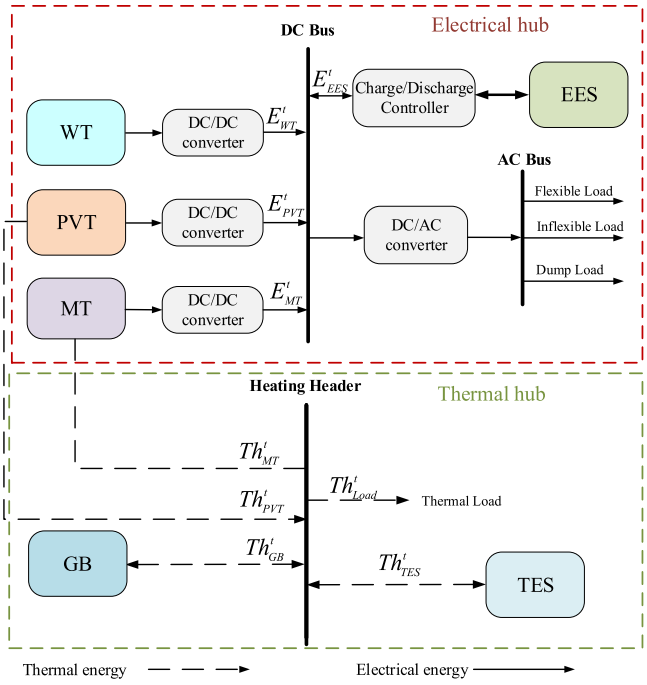


Fig. 2. SMCMG system configuration proposed in this study.

2.1. Photovoltaic-thermal panel

An increase in PV cell temperature leads to a negative effect on its electrical efficiency. It is shown that proper circulation of water through the collecting pipes of the PVT panel can extract heat and consequently keep the electrical efficiency at higher levels [26]. This extracted thermal energy can be used for meeting thermal loads. The generated thermal energy Th^t_{PVT} by the PVT and its corresponding thermal efficiency η^t_{th} vary as a function of solar irradiation G^t (kW/m^2), and can be calculated as follows [23,24],

$$Th^t_{PVT} = A_{PVT} \times F_R\left[\tau\alpha_{PVT} \times G^t - U_{loss} \times \left(T^t_f - T^t_a\right)\right] \quad (1)$$

$$\eta^t_{th} = F_R\left[\tau\alpha_{PVT} - \frac{U_{loss} \times \left(T^t_f - T^t_a\right)}{G^t}\right] \quad (2)$$

where A_{PVT} is the area of the panel (m^2), and $U_{loss}\tau\alpha_{PVT}$, and F_R are three PVT parameters and denote the heat loss coefficient, transmission and absorption coefficient, and heat removal efficiency of the PVT panel, respectively. T^t_f and $T^t_a$are, respectively, the inlet water and the ambient temperatures. The electrical efficiency $\eta^t_E$of the PVT can be obtained as follows:

$$\eta^t_E = \eta_{ref} \times \left(1 - \beta_{ref} \times \left(T^t_M - NOCT\right)\right) \quad (3)$$

where constants η_{ref}, β_{ref}, and $NOCT$ are the reference efficiency, temperature coefficient (K^{-1}), and nominal operating temperature of the PVT cell, respectively. $T_M$is also determined through the following equation.

$$T^t_M = T^t_f + \left(Th^t_{PVT} \times (1 - F_R)\right) \Big/ (A_{PVT} \times F_R \times U_{loss}) \quad (4)$$

2.2. Wind turbine

The hourly power generated by a WT (E^t_{WT}) at the wind speed $V^t_w$is approximated by Ref. [24].

$$E^t_{WT} = \begin{cases} 0 & V^t_w < V_{w,cut-in} \,||\, V^t_w > V_{w,cut-out} \\ V_w^{t\,3} \times \left(\frac{E_{WT,r}}{V^3_{w,r} - V^3_{w,cut-in}}\right) - E_{WT,r}\left(\frac{V^3_{w,cut-in}}{V^3_{w,r} - V^3_{w,cut-in}}\right) & V_{w,cut-in} < V^t_w < V_{w,r} \\ E_{WT,r} & V_{w,r} < V^t_w < V_{w,cut-out} \end{cases} \quad (5)$$

where parameters $V_{w,cut-in}$, $V_{w,r}$, and $V_{w,cut-out}$ represent the cut-in, rated, and cut-out speeds, respectively, and $E_{WT,r}$ signifies the rated power of the WT. The wind speed ($V^t_{w,2}$) at the desired hub height $h_2$can be calculated from the measured data at the anemometer heighth_1, as follows [6]:

$$\frac{V^t_{w,2}}{V^t_{w,1}} = \left(\frac{h_2}{h_1}\right)^{\alpha} \quad (6)$$

where, $\alpha$is the friction coefficient of the landscape.

2.3. Micro-turbine

The hourly fuel consumption of an MT (FU_{MT}^{t}) is a function of its output powerE_{MT}^{t}, and determined by the following equation [27],

$$FU_{MT}^{t} = (E_{MT}^{t})/(HV \times \eta_{PM}^{t}) \tag{7}$$

where HV signifies the heat value of the fuel (kWh/m^3). The electrical efficiency η_{PM}^{t} is affected by the number of paralleled MTs and their rated power $E_{MT,r}$, calculated at each part-load by

$$\eta_{PM}^{t} = \left(0.8838\Phi^{t^3} - 2.182\Phi^{t^2} + 2\Phi^{t} + 0.29\right).\eta_{MT,r} \tag{8}$$

$$\Phi^{t} = (E_{MT}^{t})/(N_{MT}.E_{MT,r}) \tag{9}$$

and the net recovered thermal energy produced by each MT (Th_{MT}^{t}) is, as follows:

$$Th_{MT}^{t} = E_{MT}^{t}/(\eta_{PM}^{t}) \times \eta_{MT}^{Th} \tag{10}$$

where $\eta_{MT}^{Th}$is the thermal efficiency of the MT.

2.4. Electrical energy storage

The charge level of the EES at each time interval (SOC_{EES}^{t}) is a function of the charge level of the EES at the previous time interval (SOC_{EES}^{t-1}), incoming/outgoing powerE_{EES}^{t}, and its self-discharge rate σ, and obtained using (11) and (12), respectively.

$$SOC_{EES}^{t} = SOC_{EES}^{t-1} \times (1-\sigma) - E_{EES}^{t} \times \eta_{EES}^{char} \tag{11}$$

$$SOC_{EES}^{t} = SOC_{EES}^{t-1} \times (1-\sigma) - E_{EES}^{t} \Big/ \eta_{EES}^{dech} \tag{12}$$

where $\sigma\eta_{EES}^{char}$, and η_{EES}^{dech} symbolize the hourly self-discharge rate, charging efficiency, and discharging efficiency of the EES, respectively. In the above equations, negative values of E_{EES}^{t} denote charging and positive values signify discharging modes. The total capacity of the battery bank Cs_{EES} is computed based on the number of batteries $N_{EES}$and the nominal capacity of the battery$S_{EES,r}$,

$$Cs_{EES} = S_{EES,r} \times N_{EES} \tag{13}$$

Over the operation time, the performance of the battery deteriorates and its capacity drops. Different factors such as calendar, operating conditions, and the number of cycles can affect this issue [28]. In this regard, various models based on the operating condition of the battery have been developed to fit the state of health of the battery [29]. In this study, as in the designing phase and future operation of the battery is indeterminate, a linear model is considered (see Fig. 3) so that the remaining capacity Cs_{EES}^{cycle} is updated according to the number of cycles ($cycle$) using

$$Cs_{EES}^{cycle} = Cs_{EES} - \frac{Cs_{EES} - Cs_{EES}^{Min}}{Cycle^{Max}} \times cycle \tag{14}$$

where Cs_{EES}^{Min} and $Cycle^{Max}$represent the end of life capacity and estimated maximum cycle count of the EES.

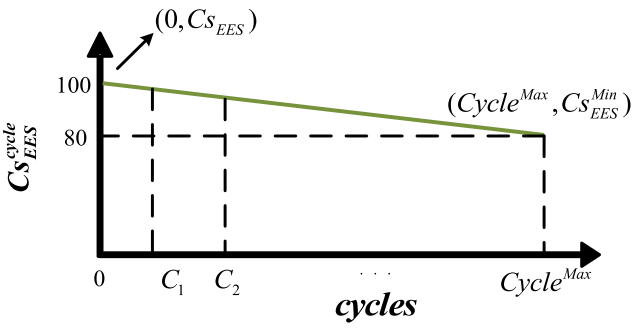


Fig. 3. Remaining capacity of the EES based on the number of cycles.

2.5. Thermal energy storage

The stored energy in TES at each time interval (SOC_{TES}^{t}) can be calculated, as follows [30]:

$$SOC_{TES}^{t} = SOC_{TES}^{t-1} \times (1-\varepsilon) - Th_{TES}^{t} \Big/ \eta_{TES} \tag{15}$$

where ε represents the heat loss coefficient (h^{-1}), and Th_{TES}^{t} is the output power of the TES, and its positive and negative values identify discharging and charging modes of TES, respectively.

2.6. Supplementary boiler

As the heat backup unit, the GB fuel consumption (FU_{GB}^{t}) is a function of its produced thermal energy $Th_{GB}^{t}$and efficiency ($\eta_{GB,r}$), obtained as below [27]:

2.7. Energy management system

The EMS administrates the SMCMG for proper energy conversion, distribution, and storage. In general, two main strategies exist for the operation of the EMS in SMCMGs; rule-based and optimization-based. The rule-based operation strategy, which requires low computational burden, is suitable for SMCMGs with a limited number of dispatchable micro-sources. But, for complex energy systems with several dispatchable micro-sources, the rule-based operation strategy needs a large number of rules to be devised, while handling all constraints of the SMCMG is not possible. Thus, the optimization-based operation strategy is used for such systems, however it entails much higher computation time and implementation cost.

In the SMCMG considered in this study, a rule-based operation strategy is devised based on giving priority to RE sources, such that thermal and electrical loads are fulfilled while maximum RE utilization and environmental considerations along with fast computation time are satisfied simultaneously (see Fig. 4). It is noted that as directed by EMS, 100% of thermal load is always met, however for the electrical load, this matter is controlled according to the pre-specified reliability level ($LPSP^{\max}$).

3. Problem formulation

The optimal sizing of an SMCMG system would determine the optimal number of system components and their capacities based on techno-economic considerations. This problem is intrinsically modeled as a nonlinear optimization problem with the following objective cost function and constraints.

3.1. Objective cost function

In the present study, the economic analysis is conducted based

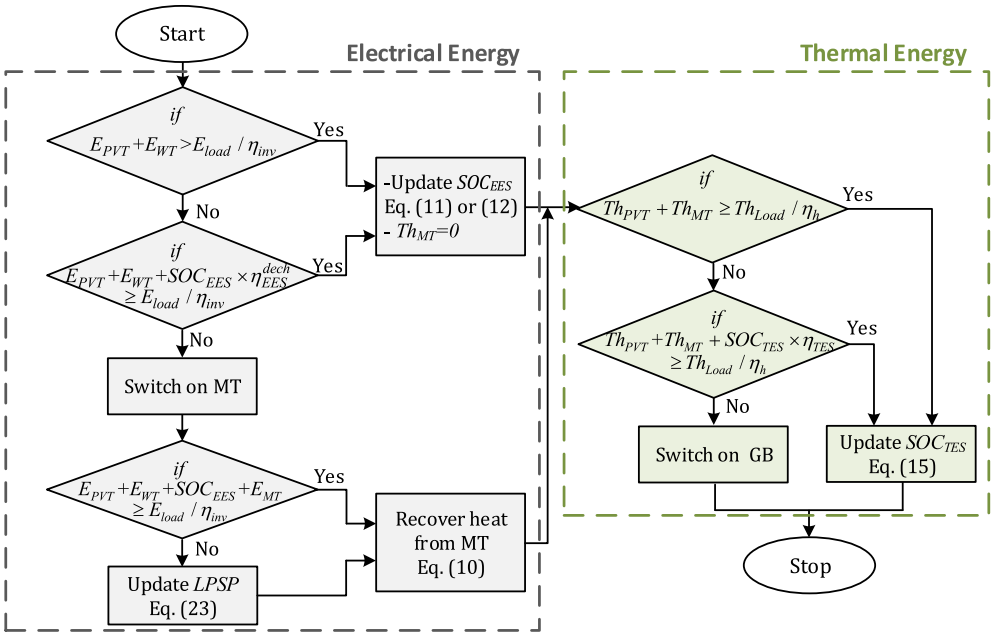


Fig. 4. Flowchart of EMS

on TAC which includes investment and operation and maintenance costs, as follows [24]:

$$Min\{TC_{Cap}+TC_{O\&M}\} \tag{17}$$

where TC_{Cap} is system capital cost and includes investment and replacement costs, and $TC_{O\&M}$is system maintenance cost which covers fuel, operation, and maintenance costs.

$$\begin{aligned}TC_{Cap}=\ & CRF\Big\{N_{WT}\times \mathrm{Pr}_{WT}+A_{PVT}\times \mathrm{Pr}_{PVT}+E_{MT,r}\ \times N_{MT}\times \mathrm{Pr}_{MT}+\mathrm{Pr}_{EES}^{P}\times Cs_{EES}\\ &+Cs_{TES}\times \mathrm{Pr}_{TES}+\mathrm{Th}_{GB,r}\times \mathrm{Pr}_{GB}+\mathrm{Pr}_{Conv/inv}^{P}\times N_{Conv/Inv}\Big\}\end{aligned} \tag{18}$$

$$\begin{aligned}TC_{O\&M}=\ & N_{WT}\times E_{WT,r}\times C_{WT}^{O\&M}\\ &+\sum_{t=1}^{8760}\left(\begin{matrix}E_{PVT}^{t}\times\ C_{PVT}^{O\&M}+E_{MT}^{t}\ \times C_{MT}^{O\&M}+E_{EES}^{t}\times\ C_{EES}^{O\&M}+Th_{TES}^{t}\times C_{TES}^{O\&M}\\ +Th_{GB}^{t}\ \times C_{GB}^{O\&M}+(FU_{MT}+FU_{GB})\times\ C_{fuel}\end{matrix}\right)\end{aligned} \tag{19}$$

where Pr_{WT}, Pr_{PVT}, Pr_{MT}, Pr_{TES}, and Pr_{GB} denote the prices per unit of WT ($/number), PVT ($/M^2), MT ($/number), TES ($/kWh), and GB ($/kW), respectively, and $C_{WT}^{O\&M}$, $C_{PVT}^{O\&M}$, $C_{MT}^{O\&M}$, $C_{EES}^{O\&M}$, $C_{TES}^{O\&M}$and$C_{GB}^{O\&M}$signify the operation and maintenance costs associated with WT, PVT, MT, EES, TES, and GB, respectively. N_{MT} ,N_{WT} and $N_{Conv/Inv}$ are the numbers of MTs. WTs, and power electronic devices, respectively, $Cs_{TES}$is the capacity of the TES (kWh), $\mathrm{Th}_{GB,r}$ is the rated power of the GB, and C_{fuel} is the fuel price ($/m^3).

It is to be noted that in the above equations, N_{WT}, A_{PVT}, N_{MT}, Cs_{EES}, Cs_{TES}, $Th_{GB,r}$ as well as E_{PVT}^{t}, E_{MT}^{t}, E_{EES}^{t}, Th_{TES}^{t}, Th_{GB}^{t}, FU_{MT}^{t}, $FU_{MT}^{t}$are variables and determined through the simulation, and the rest are given parameters.

The capital recovery factor (CRF) is used to calculate the annual equivalent of capital cost based on initial capital cost, over a specified time (n) and the interest rate of i [2].

$$CRF=\frac{i(i+i)^{n}}{(1+i)^{n}-1} \tag{20}$$

The lifetimes (n′) of EES and converter/inverter are lower than the life span of the SMCMG (n). By means of single payment present worth factor, the present worth of EES Pr_{EES}^{p} and converters/inverters $\mathrm{Pr}_{\mathrm{Conv/inv}}^{p}$ can be calculated using [24],

$$\mathrm{Pr}_{EES}^{p}=\mathrm{Pr}_{EES}\left(1+\sum_{k}\frac{1}{(1+i)^{k}}\right)\qquad k=n',2n',\ldots,Kn'\leq n \tag{21}$$

$$\mathrm{Pr}_{\mathrm{Conv/inv}}^{p}=\mathrm{Pr}_{\mathrm{Conv/inv}}\left(1+\sum_{k}\frac{1}{(1+i)^{k}}\right)\qquad k=n',\ 2n',\ldots,Kn'\leq n \tag{22}$$

where Pr_{EES} and $\mathrm{Pr}_{\mathrm{Conv/inv}}$ are prices per unit of EES and converters/inverters, respectively.

3.2. Constraints

3.2.1. Reliability index

The reliability of the SMCMG is evaluated in term of LPSP, which is defined as the long-term average loss of electrical load. An LPSP of zero represents that the entire electrical loads of the SMCMG are met, and 1 signifies that the loads are not met. For a specified period of τ (8760 h of a year), the LPSP indicator is computed based on a cumulative update of loss of the power supply (LPS) when the available hourly electrical energy is lower than the load, as follows [31]:

$$LPSP = \frac{\sum_{t=1}^{\tau} LPS^t}{\sum_{t=1}^{\tau} E_{Load}{}^t} \tag{23}$$

The inequality constraint that must be respected in the optimization procedure is:

$$LPSP \leq LPSP^{\max} \tag{24}$$

where $LPSP^{\max}$is a specified parameter and depends on the application of the SMCMG.

3.2.2. Storage systems

For charging/discharging of TES and EES, the following constraints must be satisfied in each hour.

$$SOC_{TES}^{\min} \leq SOC_{TES}^{t} \leq Cs_{TES} \tag{25}$$

$$SOC_{EES}^{\min} \leq SOC_{EES}^{t} \leq Cs_{EES} \tag{26}$$

$$\mathrm{SOC}_{\mathrm{EES}}^{\min} = (1 - \mathrm{DOD}) \times Cs_{EES} \tag{27}$$

where DOD denotes the depth of discharge.

3.2.3. Thermal power

Following constraint ensures that the SMCMG system supplies the thermal loads Th_{Load}^{t}in each time interval.

$$Th_{Load}^{t} \Big/ \eta_h \leq Th_{PVT}^{t} + Th_{MT}^{t} + Th_{TES}^{t} + Th_{GB}^{t} \tag{28}$$

where $\eta_h$represents the heat use efficiency of the SMCMG.

3.2.4. Boundary limits of optimization variables

In this study, the optimization variables (OVs) are-A_{PVT},N_{WT},N_{MT},N_{EES},Cs_{TES} and $Th_{GB,r}$. The first five OVs are offered by the optimization algorithm through a stochastic procedure in each iteration, which must be within the allowable limits considering Eq. (29).

$$OV = \begin{cases} OV^{\min} & if\ OV < OV^{\min} \\ OV^{\max} & if\ OV > OV^{\max} \\ OV & otherwise \end{cases} \tag{29}$$

and, after a one-year simulation for each offered solution,$Th_{GB,r}$ can be found by the following.

$$Th_{GB,r} = \max\left(Th_{GB}^{t}\right) \tag{30}$$

Note that $E_{MT}^{t}Th_{MT}^{t}$, E_{EES}^{t}, Th_{TES}^{t}, and Th_{GB}^{t} at each time interval, are determined by the rules of EMS strategies according to optimum capacities of the micro-sources and hourly load profile and are not determined through an optimization procedure, as shown in Fig. 5. This figure illustrates the SMCMG implementation and TAC and LPSP calculation procedure. The simulation starts with the initialization of OVs from the optimization algorithm. Then, based on the hourly meteorological data, power generations of PVT and WTs are calculated, and the SMCMG is executed according to the defined EMS (Fig. 4). At last, the LPSP and TAC of the SMCMG based on the initialized OVs are determined.

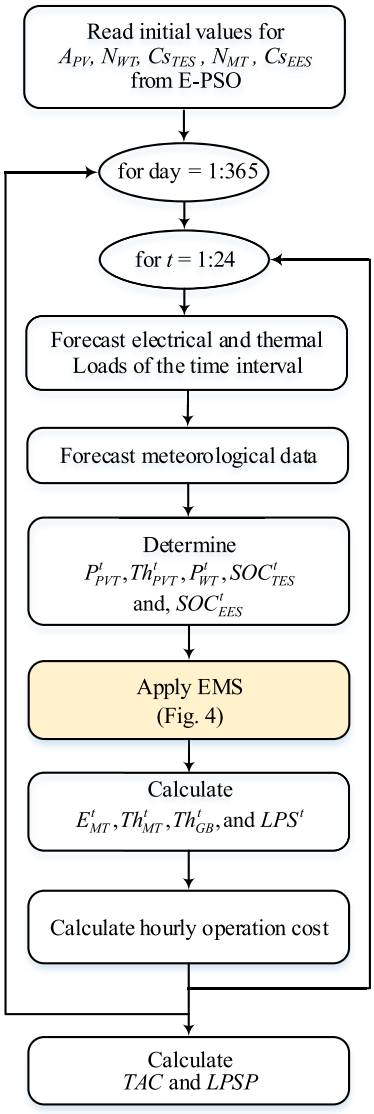


Fig. 5. Flowchart of SMCMG implementation.

4. Optimization method

Optimal sizing of the SMCMG is a complex optimization problem and thus employment of a powerful algorithm to ensure the reliability of the results is crucial. Metaheuristic optimization algorithms play a significant role in modern optimization. PSO algorithm is among the most popular evolutionary algorithms and has been used in almost all areas of optimization [32]. The main two features of this algorithm are its relatively good performance and fast convergence. The E-PSO algorithm employed in this study is based on the enhancement of the PSO algorithm. This algorithm

has been recently developed by authors and applied for the complex and nonconvex optimization problems of economic dispatch for large-scale power system [33] and optimal integration of renewable energy into CCHP [24] and CHP [25] systems, and its robustness and superiority among various optimization techniques have been thoroughly proved. This algorithm is described below.

4.1. Particle swarm optimization algorithm

This algorithm imitates flocks of birds searching for food [34,35]. For *nPop*birds identified as particles distributed in the *DM* dimensional search space, position X_j^{it} and velocity $V_j{}^{it}$at iteration *it* are determined based on the corresponding fitness value$f_{X,j}(j = 1,2, \ldots, nPop)$, best position$Pbest_j^{it}$, and the global best location of the entire swarm*Gbest*, as follows:

$$\begin{cases} V_j^{it+1} = \left(\omega V_j^{it} + C_1.\, Rand_1().\left(Pbest_j^{it} - X_j^{it}\right) + C_2.\, Rand_2().\left(Gbest\ - X_j^{it}\right)\right) \\ X_j^{it+1} = X_j^{it} + V_j^{it+1} \end{cases} \tag{31}$$

where, $Rand_1$ and $Rand_2$ symbolize the random numbers from the uniform distribution in the range of [0, 1]. Also,ω (inertia weight), C_1 (cognitive weight), and C_2 (social weight) are user-specified constants. Then, the velocity of each particle is tuned using Eq. (32).

$$V_j^{it+1} = \begin{cases} V_j^{it+1,\min} & if\ \ V_j^{it+1} < V_j^{it+1,\min} \\ V_j^{it+1,\max} & if\ \ V_j^{it+1} > V_j^{it+1,\max} \\ V_j^{it+1} & otherwise \end{cases} \tag{32}$$

where $V_j^{it,\min}$and $V_j^{it,\max}$are the indications of the minimum and maximum velocity limits of the particles, respectively.

4.2. Proposed evolutionary particle swarm optimization (E-PSO)

Despite the benefits of the PSO, the major shortcoming of this algorithm is its slender capability to escape from local optimum solutions and maintain the diversity of swarm [36]. Therefore, in this study, three evolutionary operators are integrated into the PSO algorithm, and the proposed algorithm is called E-PSO algorithm. Moreover, three coefficients, namely ω, C_1, and C_2, are dynamically tuned using devised arc-tangent based relations so as to enhance the searching capability of the algorithm.

4.2.1. Mutation operator

In each iteration, the mutation operator engenders the mutant vector (M_j^{it}) of each target vector (X_j^{it})using the following equation:

$$M_j^{it} = X_{r_1^j}^{it} + F.\left(X_{r_2^j}^{it} - X_{r_3^j}^{it}\right) \qquad j = 1,2,\ldots,nPop \tag{33}$$

where indices r_1^j, r_2^j and $r_3^j$are random integers within the interval of [1,*nPop*]. *F* is a random number within the range of $[F^{\min}, F^{\max}]$.

4.2.2. Crossover operator

At this stage, the crossover operator generates the associated trial vector (T_j^{it}) of the $X_j^{it}$and M_j^{it}, as follows:

$$if \begin{cases} X_j^{it} = \left(x_j{}^1, x_j{}^2 \ldots x_j{}^{DM}\right), \\ M_j^{it} = \left(m_j{}^1, m_j{}^2 \ldots \mathrm{m}_j{}^{DM}\right), \quad \text{Then} \quad t_j{}^b \\ T_j^{it} = \left(t_j{}^1, t_j{}^2 \ldots t_j{}^{DM}\right), \end{cases}$$
$$= \begin{cases} m_j{}^b \quad if\ (rand_b[0,1) \le CR)\ or (\mathrm{b} = \mathrm{b}_{rand}) \\ x_{j,k}{}^b \quad otherwise \\ \mathrm{b} = 1,2,\ldots,DM \end{cases} \tag{34}$$

where,$b_{rand}$is a random integer in the range of [1, *DM*], *CR* is a constant in the interval of [0,1). In general, the crossover operator copies the bth element of the $M_j{}^{it}$as the corresponding element in T_j^{it} if$(rand_b[o,\ 1) \le CR)$or$b = b_{rand}$. Otherwise, it copies that element fromX_j^{it}.

4.2.3. Selection operator

After the crossover operator, the selection operator by comparing $X_j{}^{it}$and the T_j^{it}, elects the one with better fitness value to compose the population of the next searching stage.

4.2.4. Dynamic tuning of parameters

Basically, ω, C_1, and $C_2$are the three user-specified constants that strongly influence the effective searching capability of the E-PSO algorithm. Based on Eq. (31), it is inferred that a large ω leads to better global searching in the search space, while the small ω improves the local search capability of the E-PSO. C_1 stresses the impact of the personal best experience on particle velocity. Therefore, in the beginning, C_1 should be large enough to enhance the global searching and escaping from local optimum. However, during the final iterations, C_1 should be small to realize the local search. C_2 signifies the influence of global best experience. Therefore, at the start of the searching process, it should have little influence on particles velocity so as to enhance the local search. Then, as the iteration proceeds, this parameter should gradually increase to improve the local search. Therefore, it is expected that the dynamic tuning of these parameters using the following equations can strongly improve the performance of the algorithm.

$$\omega(it) = 0.5 \times \left(\omega^{\min} + \omega^{\max}\right) + K_\omega \times \left[\arctan\left(\frac{-2\pi}{MaxIt} \times it + \pi\right)\right] \times \left(\omega^{\max} - \omega^{\min}\right) \tag{35}$$

$$C_1(it) = 0.5 \times \left(C_1{}^{\min} + C_1{}^{\max}\right) + K_{C_1} \times \left[\arctan\left(\frac{-2\pi}{MaxIt} \times it + \pi\right) \times\right] \times \left(C_1{}^{\max} - C_1{}^{\min}\right) \tag{36}$$

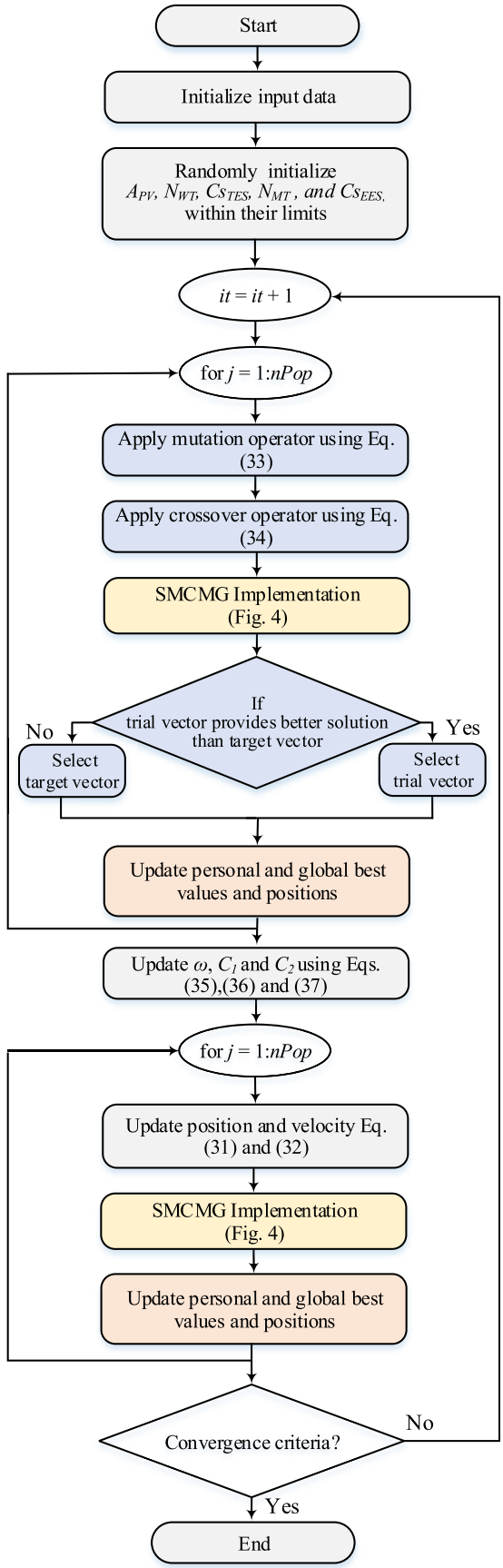


Fig. 6. Flowchart of the developed procedure.

$$C_2(it) = 0.5 \times \left(C_2^{\min} + C_2^{\max}\right) + K_{C_2} \times \left[\arctan\left(\frac{2\pi}{MaxIt} \times it - \pi\right)\right] \times \left(C_2^{\max} - C_2^{\min}\right) \tag{37}$$

where K_ω, K_{C_1}, and K_{C_2} are shape-shift coefficients, $\omega^{\min}$, $C_1^{\min}$, and $C_2^{\min}$ are minimum of inertia, cognitive, and social weights, and $\omega^{\max}$, $C_1^{\max}$, and $C_2^{\max}$ are minimum of inertia, cognitive, and social weights, respectively.

The flowchart of the algorithm is shown in Fig. 6. The process starts with initialization. Then, The E-PSO algorithm engenders a solution based on the defined evolutionary operators. After that, the SMCMG is formed (Fig. 5) and subsequently operated using the defined EMS (Fig. 4). Next, the TAC and LPSP of the SMCMG corresponding to this solution (OVs) are obtained. This process is repeated for all the initial population and meanwhile, the personal and global bests of the particles are updated. Afterwards, the position and velocity of the population are updated using the defined dynamic coefficient tuning, so as to compose the new population and thus solution. The noted procedure regarding the implementation of the SMCMG and calculation of the TAC and LPSP are repeated for the entire population. This process iterates until the convergence criterion is satisfied.

5. Results and discussion

5.1. System setup

The planning horizon and useful lifespan of the SMCMG are assumed to be 25 years. This long-term simulation approach allows for taking into account the intermittent nature of RE sources during days and seasons [3,25]. The hourly meteorological data of Rafsanjan city, Iran are used for simulation as shown in Fig. 7. The meteorological data and yearly load profile remain constant over the planning horizon [24,25,36]. The components specifications of the SMCMG and the parameters of E-PSO algorithm are given in Table 1 and Table 2, respectively. EES and TES are assumed to be initially empty and $LPSP^{\max}$ is set to 0.02. Regarding optimization constraints presented in Section 4, Eq. (24) is formulated as a penalty function, and the value of the penalty function is proportional to the square of excess level for LPSP and a coefficient determined by try and error. The other constraints such as stored energy in EES and TES provided power from dispatchable microsources, etc. are satisfied during the coding and implementation of the EMS of the SMCMG. The simulations are performed in MATLAB environment.

5.2. Validation of optimization results

It is of great importance to validate the performance of the E-PSO algorithm and the reliability of the simulation results. In this regard, due to the stochastic nature of metaheuristic optimization algorithms, the SMCMG is simulated for 30 times by each of the widespread metaheuristic optimization algorithms as well as the E-PSO, and their results are analyzed and compared in Table 3. It is observed that the E-PSO has found a better solution (lower TAC) than the other studied algorithms. Further, the average values of the results of the E-PSO are also lower than the corresponding value of all other studied algorithms. It is worth mentioning that the worst result (highest TAC) among the 30 found results by the E-PSO is also more desirable than the best results (lowest TACs) obtained by the other metaheuristic optimization algorithms. Furthermore,

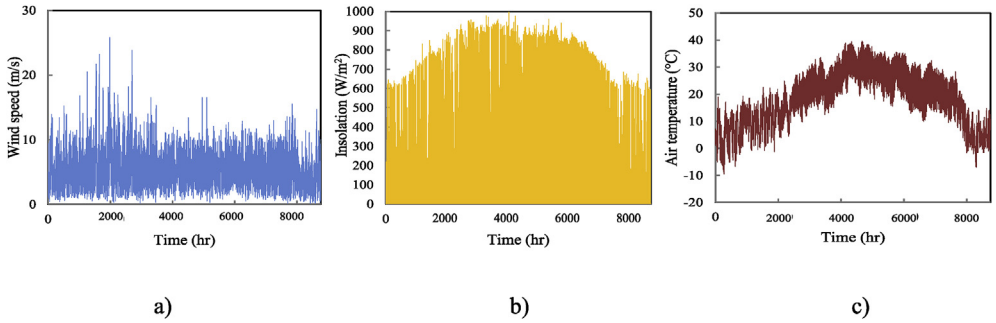


Fig. 7. Hourly profile of meteorological data during a year: a) wind speed (at the height of 10 m), b) insolation, and c) air temperature [6].

Table 1
SMCMG components specifications [10,23–25,30,37].

Parameter	Value (Unit)	Parameter	Value (Unit)
PVT		**EES**	
Pr_{PVT}	400 ($/m^2)	Pr_{EES}	130 ($/kWh)
$C^{O\&M}_{PVT}$	0.01 ($/kWh)	$C^{O\&M}_{EES}$	0 ($/kWh)
n'	25 (years)	$S_{EES,r}$	1.35 (kWh)
η_{ref}	0.15	n'	5 (years)
β_{ref}	0.005	η^{char}_{EES}, η^{dech}_{EES}	93 (%)
$\tau\alpha_{PV}$	0.75	DOD	80 (%)
U_{loss}	7 (W/m^{2}K)	σ	0.0002
F_R	0.85	Cs^{Min}_{EES}	80%
$NOCT$	297 (K)	$Cycle^{Max}$	650
WT		TES	
Pr_{WT}	10000 ($)	Pr_{TES}	33 ($/kWh)
$C^{O\&M}_{WT}$	100 ($/kW)	$C^{O\&M}_{TES}$	0 ($/kWh)
n'	25 (years)	n'	25 (years)
$V_{w,cut-in}$	2.5 (m/s)	η_{TES}	0.9
$V_{w,r}$	9.5 (m/s)	ε	0.05
$V_{w,cut-in}$	40 (m/s)	MT	
$E_{WT,r}$	5 (kW)	Pr_{MT}	2000 ($/kW)
α	0.14	$C^{O\&M}_{MT}$	0.01 ($/kWh)
Boiler		n'	25 (years)
Pr_{Boiler}	90.18 ($/kW)	$\eta_{MT,r}$	0.29
$C^{O\&M}_{Boiler}$	0.0027 ($/kWh)	η^{Th}_{MT}	0.48
n'	25 (years)	$E_{MT,r}$	30 (kW)
$\eta_{boiler,r}$	0.8	Inverter/Converter	
PV		$Pr_{Conv/inv}$	2000 ($)
Pr_{PV}	380 ($/m^2)	η_{inv}	0.95
$C^{O\&M}_{PV}$	0.01 ($/kWh)	n'	10 (years)
n'	25 (years)	Other	
η_{ref}	0.15	η_h	0.9
$NOCT$	297 (K)	C_{fuel}	1 ($/m^3)
β_{ref}	0.005	n	25 (years)

Table 2
Simulation parameters of E-PSO algorithm.

Parameter	Value	Parameter	Value
DM	5	K_{C_1}	0.4
$MaxIt$	200	C^{Max}_2	1.5
ω^{max}	1	C^{Min}_2	0.1
ω^{min}	0.4	K_{C_2}	0.4
k_ω	0.4	F^{Min}	0.2
C^{Max}_1	1.5	F^{Max}	0.7
C^{Min}_1	0.1	CR	0.4

the standard deviation of the results of the E-PSO is $9.39, which is significantly lower than those of other algorithms. Also, the lowest and highest values of TAC obtained by the E-PSO out of 30 times simulations are close to the average value of the results, so that the robustness of the E-PSO algorithm can be inferred. All of the above issues demonstrate the competence of the E-PSO algorithm, effectiveness of the devised enhancements, and reliability of the sizing results. Additionally, the results clarify that the E-PSO algorithm entails lower computation time because of its lower initial population ($nPop$) and thus lower number of cost function evaluation through the optimization procedure. For more illustration, the convergence characteristics of the studied optimization algorithms are depicted in Fig. 8. It is observed that the E-PSO algorithm can effectively proceed and the objective function continuously decreases. All of the above-mentioned considerations reveal the superiority of the E-PSO algorithm in the complex optimal problem of optimal sizing of the SMCMG.

Furthermore, Table 4 sheds light on the effectiveness of the integration of the devised enhancements to the E-PSO algorithm in simulation results. The results show that the E-PSO without the noted three evolutionary operators or non-linear tuning of C_1, C_2 and ω does not outperform other widespread evolutionary algorithms. In other words, the devised enhancements must be integrated, such that their individual benefits are simultaneously realized in the combined framework.

5.3. Operational performance of SMCMG

The optimal values for OVs, TAC, and LPSP of the SMCMG based on the E-PSO algorithm over scenarios of S1 and S2 are given in Table 5. In S1, the degradation of the EES is not considered. The optimal TAC and LPSP based on S2 are $90867.74 and 1.9987%. It is observed that with the consideration of battery degradation (S2), the optimal capacity of the battery and the TAC of the SMCMG increases, and the LPSP is also lower than in S1.

Furthermore, for more effective delivery of merits of utilization of PVT panel in the SMCMG, the hourly thermal, electrical, and total efficiencies of the PVT panel during the mid-season day of each season are illustrated in Fig. 9. It is depicted that the total efficiency (electrical efficiency plus thermal efficiency) of the PVT panel is at least 52% in summer, while in winter, the total efficiency approaches 63%. It means that as the environment temperature decreases, the total efficiencies of the PVT panel increases.

The performance of electrical and thermal hubs of the SMCMG for mid-season days of each season are shown in Figs. 10 and 11, respectively. Please note that the upper part of the abscissa axis designates the generated (injected) energy, while the lower part indicates the drawn powers (by loads or storage systems) at each time interval. Fig. 10 shows that the simultaneous utilization of PVT and WT results in alleviating their inherent intermittencies. It is

Table 3
Validation of E-PSO algorithm for optimization of SMCMG out of 30 independent runs.

Optimization algorithm	Initial population	Lowest TAC ($)	Average TAC ($)	Highest TAC ($)	Median TAC ($)	Standard deviation	Simulation time (s)
DE	40	90889.82	90952.59	91082.66	90950.18	55.18	86
PSO	45	90904.68	91147.55	91668.97	91136.83	170.21	91
GA	45	90899.32	91140.23	91685.54	91168.02	260.91	93
HSA	45	90907.67	91160.70	91759.65	91062.45	262.94	92
E-PSO	20	90867.74	90870.87	90899.04	90867.74	9.39	81

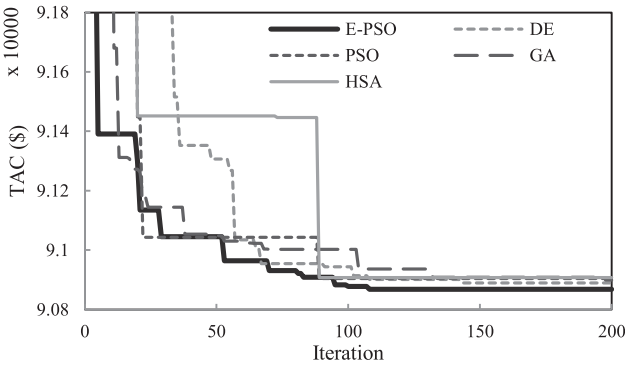


Fig. 8. Convergence behavior of studied algorithms for optimal SMCMG sizing.

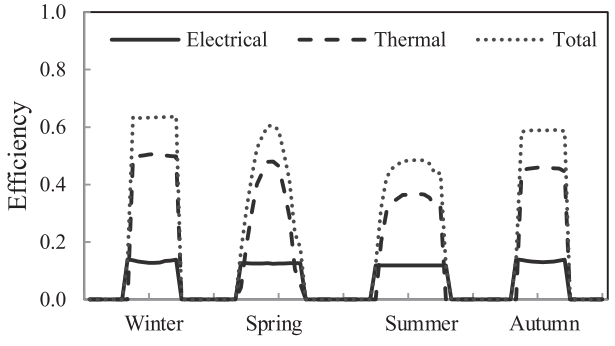


Fig. 9. PVT daily electrical and thermal efficiencies during the mid-season day of each season.

observed, when there is no PVT panel power generation, WTs, EES, or MT meets the electrical base load, and TES, MT or GB satisfies the thermal loads for the SMCMG. During the noon hours, RE sources play the fundamental role, especially PVT panel which provide the electrical and thermal loads. It is also observed that in every season, the high electrical and thermal efficiency of the PVT panel results in energy generation above the load levels to allow storage by EES and TES. It is further observed that MT and GB are activated during morning and night however, the backup GB unit is switched on only during winter and autumn. During spring and summer, the thermal energy of PVT, TES, and MT is sufficient, and limited surplus energy may be dissipated in the outdoor air. Moreover, in this regard, limited load shedding may occur with respect to the pre-specified $LPSP^{max}$ for the SMCMG.

5.4. Different configuration alternatives of SMCMG

To justify the competence of the proposed configuration, different configuration alternatives for the SMCMG are studied and compared with the proposed SMCMG with respect to economic feasibility, energy efficiency, and reliability considerations (see Table 6). Furthermore, the contribution of each component of the SMCMG to the operation of the SMCMG is also clarified. It is observed that, in the absence of WT, the size of the PVT is roughly doubled and TAC increases by 17.06%. Also, excluding the PVT results in an enormous increase of TAC by 35.68% and, when a PV is used instead of PVT, the TAC increases by 33.26%, emphasizing the significance of integration of PVT and WT in the SMCMG. In the absence of PVT (RE based-CHP micro-source), the number of MTs (combustion-based CHP micro-source) is doubled. This increase in the number of utilized MTs results in a level of reliability higher than required for the SMCMG, consequently leading to excessive costs. On the other hand, the exclusion of MTs leads to an augmentation of the PVT panel area and increases the storage systems' capacities as well. It is also worth mentioning that the absence of WTs results in a substantial increase in storage system capacities, but the absence of the PVT decreases the required capacities for storage systems. Moreover, between the studied configuration alternatives, a cost versus the desired level of reliability($LPSP^{max}$) comparative analysis is conducted, as depicted in Fig. 12. It is observed that the curve associated with the proposed configuration is always below the curves of other configuration

Table 4
Sensitivity of TAC results of SMCMG to E-PSO algorithm enhancements based on 30 independent runs.

Optimization algorithm	Lowest TAC ($)	Average TAC ($)	Highest TAC ($)
PSO	90904.68	91147.55	91668.97
E-PSO without mutation and crossover operators	90899.12	90942.01	91036.14
E-PSO without C_1,C_2 andωnon-linear tuning	90899.04	90929.46	90996.28
E-PSO	90867.74	90870.87	90899.04

Table 5
Obtained results for optimal sizing of SMCMG at $LPSP^{max} = 0.02$.

Scenario	A_{PV} (m^2)	N_{WT}	N_{MT}	N_{EES}	Cs_{TES} (kWh)	$Th_{GB,r}$ (kWh)	$LPSP$ (%)	TAC ($)
S1	633	11	1	210	867	128	1.9987	90454.77
S2	633	11	1	224	867	128	2	90867.74

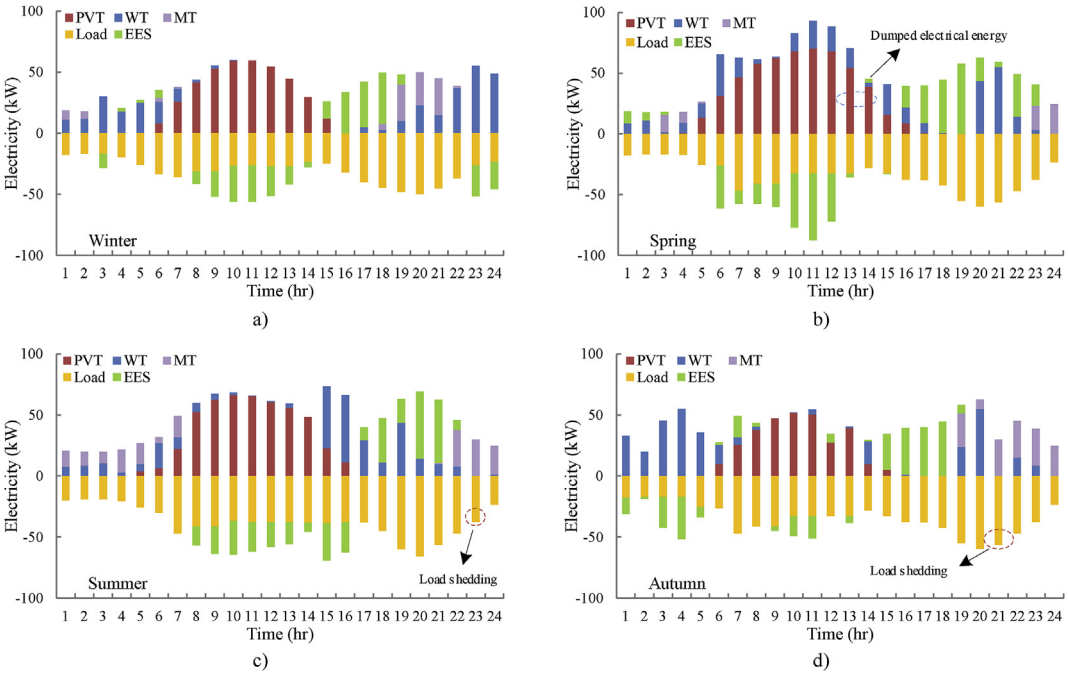


Fig. 10. SMCMG electrical performance during the mid-season day of each season.

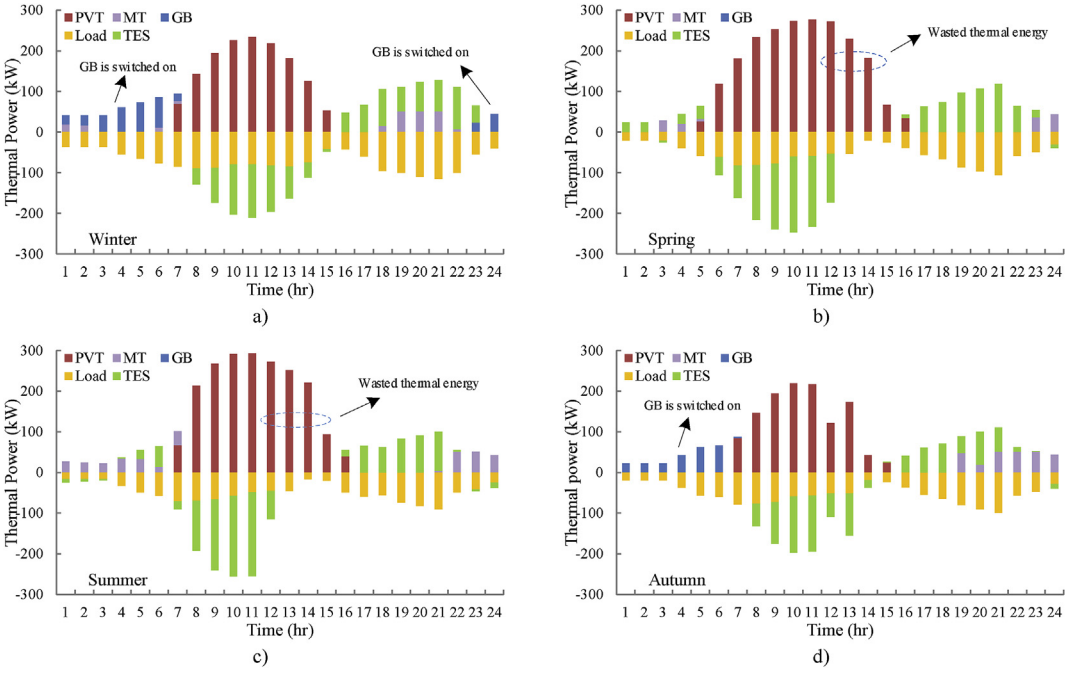


Fig. 11. SMCMG thermal performance during the mid-season day of each season.

Table 6
Comparison of performance of different SMCMG configurations for $LPSP^{max} = 0.02$.

SMCMG	SMCMG configuration	A_{PV} (m^2)	N_{WT}	N_{MT}	N_{EES}	Cs_{TES} (kWh)	$Th_{GB,r}$ (kWh)	$LPSP$ (%)	TAC ($)	Dump energy (%)	
										Electrical	Thermal
1	PVT-WT-MT-EES-TES-GB	633	11	1	224	867	128	2	90867.74	3.2	6.2
2	PV-WT-MT-EES-TES-GB	111	7	2	10	126	100	0.1005	121092.31	7.1	11.2
3	WT-MT-EES-TES-GB	N/A	11	2	100	133	128	0.0736	123294.21	8.1	11.2
4	PVT-MT-EES-TES-GB	1329	N/A	1	479	1220	77	2	106372.49	14.4	34
5	PVT-WT-EES-TES -GB	1326	8	N/A	1258	1278	128	2	113872.24	26.8	33

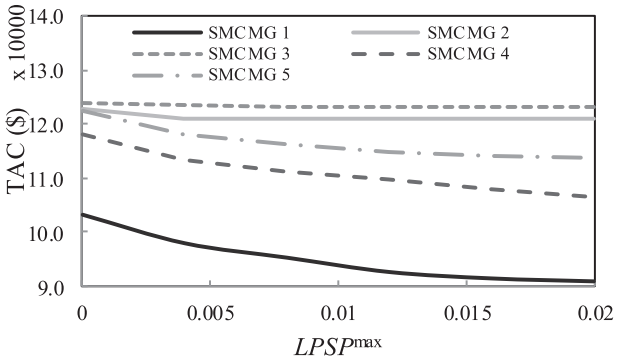


Fig. 12. Cost-reliability analysis between different SMCMG configuration.

alternatives. Further, and as expected, the total annual percentage of the dump electrical and thermal energy of the different configuration alternatives for the SMCMGs are also given in Table 6. It is demonstrated that the simultaneous integration of PVT, WTs, MT, EES, TES, and GB in an optimized framework leads to a slender percentage of dumped electrical and thermal energy, namely 3% and 6% of the electrical and thermal energy, respectively. It can be concluded that the proposed SMCMG is the most energy- and cost-efficient configuration that could be used in Rafsanjan city, Iran.

5.5. Sensitivity analysis

At this stage, sensitivity analyses are conducted on the capacity of EES (Cs_{EES}) and TES(Cs_{TES}), NG price(C_{fuel}), and interest rate (i) (see Fig. 13). The optimal value of the capacity of storage systems occurs at $Cs_{EES} = 302.4$(kWh) and $Cs_{TES} = 867$(kWh), and an increase in TAC is expected for all values greater or less than the optimal values of storage systems. Further, it is found that for values greater than the optimal value of the TES, the curve gradient is lower than the corresponding values for EES, due to the lower initial cost of TES (Fig. 13(a)). As for the impact of NG price on the economy of the SMCMG, it is realized that there is a turning point at $C_{fuel} = 0.4(\$/m^3)$, after which the utilization of RE resources and EES are not affordable beside the MT, backup GB, and TES (Fig. 12(b)). It is further inferred that the interest rate plays a significant role in the cost-effectiveness of the SMCMG (Fig. 13(c)).

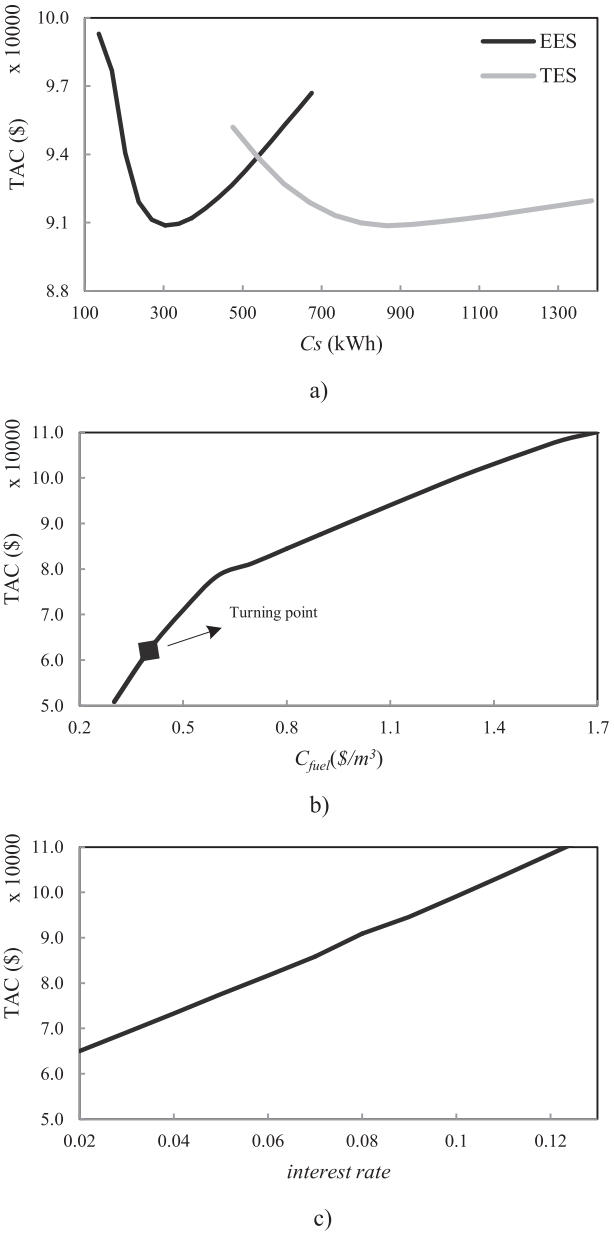


Fig. 13. Sensitivity analyses of TAC of SMCMG.

5.6. Environmental effects

To study the environmental effect of the proposed SMCMG, it is compared with the conventional energy system relying on steam power plants and NG for electricity and heat generation, respectively [23]. Table 7 lists the annual reduction of GHG emission thanks to the utilization of the proposed SMCMG.

6. Conclusions and recommendations

In this study, the optimal sizing and comprehensive techno-economic analysis of an energy- and cost-efficient SMCMG have been examined. In the proposed SMCMG, PVT, WTs, MTs, EES, TES, and GB unit have been optimally integrated. For proper energy distribution in this integrated system, a rule-based EMS has been introduced giving priority to RE utilization. For solving the optimization problem, a newly developed E-PSO algorithm has been used. This algorithm has been validated against the performance of other widespread metaheuristic optimization algorithms. It was demonstrated to find a better solution, converge faster, and display less variance than the other algorithms. The energy efficiency, economic viability and the reliability of the proposed configuration of the SMCMG have been demonstrated to be superior to other configurations through comparative analyses. Based on the simulation of various configurations of SMCMGs, it can be concluded that the utilization of the PVT panel is significant; otherwise, the cost and dump energy of the system increases by 35.68%, and 5%, respectively. For the SMCMG without the utilization of the power generation of WTs, TAC experiences a 17.06% increase. Further, it has been shown that by combining two different CHP technologies, namely PVT panel and MT, the TAC and dump energy of the SMCMG decreases by more than 25%. Moreover, it has been illustrated that

Table 7
Environmental effect of proposed SMCMG in comparison with steam power plant and natural gas.

Pollutant (kg/year)	NOx	SO_2	CO	CO_2
Steam power plant	809.89	1722.52	486.54	230286.98
NG	705.69	115.41	26.21	193126.30
MT	22.97	0.03	6.40	6930.88
GB	12.67	0.58	6.91	11571.81
Total reduction of pollutants (kg/year)	1479.94	1837.32	499.44	404910.59

using the studied SMCMG, reduces the generation of pollutants, including, most notably, the CO_2 emissions, which are decreased by 404 tonnes per year.

For future work, an electrical chiller, absorption chiller, or heat pump could be added to SMCMG for meeting local cooling loads.

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